Unix and Bash exercises

NUnix · L3 MIAGE · TD

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Exercise session

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1. Filename patterns
2. Redirections and paths
3. Scripts with arguments and interactive input
4. Search commands

Use a **practice directory** for files that you create or change.
Read commands and predict their effects before executing them.

Exercise 1: filename patterns

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List the entries in `/usr/bin` whose names:

1. Start with `m`.
2. Start with `m` and have exactly three characters.
3. Start with `m` and have at least three characters.
4. Start with `m` and have an extension after a dot.

Exercise 2: the initial files

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Directory	Contains
`practice/user1/`	`f1`, `f2`, `f3`
`practice/user2/`	`f4`

The current directory is **`practice/user2/`**.

What relative path names `user1`'s file `f1`?

Exercise 2: redirections with cat

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Using `cat` and redirections:

1. Create `f5` containing the line `5 2 3 -1`.
2. Create `f6` from `f4`, followed by `f5`.
3. Append `user1`'s `f1` to `f6`.
4. Copy `user1`'s `f2` into a new file `f7` in `user2`.

Exercise 3: a script with three filenames

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Write a script named `execution`, called with `n1 n2 n3`.

Step	Input	Program	Output
1	File `n1`	`p1`	File `n1.r1`
2	File `n2`	`p1`	File `n2.r1`
3	`n2.r1`, then `n1.r1`	`p2`	File `n3`

`p1` and `p2` read **stdin** and write **stdout**.
Concatenate the two intermediate files in the stated order.

Exercise 3: resolving the paths

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Directory	Contains
`practice/user1/`	`file1`, `file2`, `p1`, `p2`, `execution`
`practice/user2/`	Another file named `file2`

From `practice/user1/`, run:

```
./execution file1 ../user2/file2 fileN
```

Which files are read? Where do the three output files appear?

Exercise 3: interactive input

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Write another version of `execution` that asks the user for:

- The first input filename.
- The second input filename.
- The output filename.

Keep the same processing steps and concatenation order.

Exercise 4: script arguments

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Write `argument`, using `expr` to update a counter.

```
./argument one two three
```

Expected output:

```
3 arguments:
  argument 1 - one
  argument 2 - two
  argument 3 - three
```

Exercise 5: finding a pattern

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Write `mysearch`, called with a pattern and a filename:

```
./mysearch 'error' application.log
```

Print either:

- `application.log has the pattern`
- `application.log does not have the pattern`

Handle an unreadable or missing file as an **error**.

Exercise 5: repeated searches

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1. Write `mysearch2 filename`. Ask for a pattern repeatedly until the user enters **end**.
2. Write `mysearch3 f1 ... fn`. Search the supplied files for each entered pattern.

Where does each loop belong? What happens when input reaches EOF?

Exercise 6: reading find commands

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Explain each command. Identify unnecessary directory visits or ambiguous patterns.

```
find /var/log -exec grep 'www.athabasca.com' {} \;
find . -name 'rc.conf' -exec chmod o+r {} \;
find /usr/local -type f -name '*.html'
```

Exercise 6: who handles the redirection?

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Analyse this command before running it:

```
find . -name 'tp2' -exec {} < f_in > {}.out \;
```

1. Which program opens `f_in` and `{}.out`?
2. Does `find` replace `{}` inside the redirection filename?
3. How would you give each matching program its own output file?